

Final Benchmark Study Guide

PSYC 4213 Statistics · Modules 6-8 (Chapters 7-13)

Final Benchmark / Benchmark 3 covers Modules 6-8 only - NOT a comprehensive final over the whole semester. It is proctored in Canvas. The separate Final Benchmark written assignment (open book) uses Final Benchmark Data.csv; this guide focuses on the proctored multiple-choice style benchmark on Modules 6-8.

Scope reminder

- In scope: probability & sampling (Module 6), hypothesis testing (Module 7), comparing groups & relationships (Module 8).
- Out of scope: Modules 1-5 content is not retested comprehensively (only where it supports M6-8 items).
- Textbook: LSJ Chapters 7-13 (probability through one-way ANOVA intro).

Module 6 - Probability & sampling

- Basic probability rules; frequentist vs Bayesian (intro).
- Binomial and normal distributions; z-scores in probability context.
- Sampling methods; sampling error; law of large numbers; central limit theorem (conceptual).
- Point estimates; confidence intervals - interpret "we are 95% confident?" correctly.

Module 7 - Hypothesis testing

- Null (H_0) and alternative (H_a) hypotheses; one- vs two-tailed tests.
- Type I and Type II errors; alpha and power (intro).
- p-values: what they are and are not (not probability H_a is true).
- Effect sizes (Cohen's d, η^2 / ω^2 intro) alongside p-values.
- Decision rule: reject H_0 when $p < \alpha$; state conclusions in context.

Module 8 - Choosing an analysis (overview)

- Two groups, one continuous outcome -> independent or paired t-test (Welch when variances differ).
- Three or more groups, one outcome -> one-way ANOVA (F test); post-hoc comparisons.
- Two categorical variables -> chi-square test of independence; expected counts & assumptions.
- Two continuous variables -> correlation / simple regression (predict Y from X).
- Check assumptions at a "recognize the issue" level: normality, independence, homogeneity of variance.
- Match research question -> test: know when ANOVA vs t vs χ^2 vs regression is appropriate.

How to study

- Complete Concept Review and Software Practice for Modules 6-8.
- Take Final Benchmark practice test in Methods Market (36 questions per attempt).
- Review Module 7 hypothesis scenarios assignment patterns (H_0 / H_a , test choice, error types).
- For the written Final Benchmark assignment: use Final Benchmark Data.csv (Study_Mode $\times$ Score) and follow the "What to write" checklist in Assignment Help.

Methods Market: <https://methods-market.cleu.com/class/statistics> - Concept Review, practice tests, and Assignment Help for each module.